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Amyloid precursor protein as a diagnostic marker for biliary atresia

Abstract

Provided are novel diagnostic and prognostic methods for biliary atresia using amyloid precursor protein.

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Background/Summary

SEQUENCE LISTING

(1) A Sequence Listing conforming to the rules of for Application filed prior to Jul. 1, 2022 is hereby incorporated by reference. Said Sequence Listing has been filed as an electronic document via the USPTO patent electronic filing system in ASCII formatted text. The electronic document, created on May 17, 2025 is entitled "104525-1253597-000310US_ST25", and is 82,065 bytes in size.

BACKGROUND

(2) Biliary atresia is a disease of the liver and bile ducts that occurs in infants. Typically, symptoms of biliary atresia appear or develop about two to eight weeks after birth. Cells within the liver produce bile, a digestive liquid that helps digest fat. It also carries waste products from the liver to the intestines for excretion. This network of channels and ducts in which bile is produced, digests fat, and eventually moves to intestines is called the biliary system. Under the normal physiological conditions, the biliary system allows the bile to drain from the liver into the intestines. When a baby has biliary atresia, however, bile flow from the liver to the gallbladder is blocked. This causes the bile to be trapped inside the liver, which can quickly cause damage and scarring of the liver cells, resulting in liver cirrhosis and eventually liver failure.

(3) While the exact cause of biliary atresia is yet to be fully illustrated, it is believed that some infants develop biliary atresia due to their bile ducts not formed properly during pregnancy. For most other children with biliary atresia, their bile ducts may be damaged by the body's immune system in response to a viral infection acquired after birth. Although biliary atresia is a rare disease, it affects the Asian population at a notably higher rate than other ethnicities. Diagnosis of biliary atresia relies on blood tests, ultrasound of the abdomen, liver biopsy, as well as diagnostic surgery (which may include an operative cholangiogram). Once diagnosed, biliary atresia is treated by Kasai procedure or hepatoportoenterostomy. With a 65-85% success rate routinely achieved by this procedure, failure still can result in a small percentage of infants, who will then likely require a liver transplant for long term survival. Because of this life-threatening potential of biliary atresia, there exists an urgent need for new and more reliable methods for the early diagnosis of infants suspected of having biliary atresia and for risk assessment among those who have undergone Kasai operation. This invention fulfills this and other related needs.

SUMMARY OF THE INVENTION

(4) The present inventors observed increased expression of amyloid precursor protein in the liver tissue of an infant who is at risk of developing biliary atresia or is suffering from biliary atresia and has a poor outcome of biliary atresia treatment by Kasai operation. This discovery thus allows new methods to be devised for use in the diagnosis or prognosis of biliary atresia.

(5) In the first aspect, the present invention provides a method for diagnosing or assessing the risk of biliary atresia (BA) in an infant by detecting in a liver sample taken from the infant the level of amyloid precursor protein (APP). The method includes these steps: (i) determining the expression level of APP in a liver sample taken from the infant; (ii) detecting an increase in the APP expression level in comparison to a standard control level; and (iii) determining the infant as having BA or at an increased risk of developing biliary atresia.

(6) In some embodiments, the APP expression level is APP mRNA level or protein level. In some embodiments, step (i) comprises a reverse transcription polymerase chain reaction (RT-PCR) or an immunoassay. In some embodiments, the liver sample is a liver biopsy. In some embodiments, the method further includes after step (iii) a step of performing at least one additional diagnostic test for BA to confirm the diagnosis. In some embodiments, the method may further include after step (iii) a step of treating the infant with Kasai procedure.

(7) In the second aspect, the present invention provides a method for assessing the likelihood of poor outcome (e.g., ineffective Kasai procedure) in an infant after the infant has been diagnosed with BA and has received Kasai operation. The method includes these steps: (i) determining the expression level of APP in a liver sample taken from the infant; (ii) detecting an increase in the APP expression level in comparison to a standard control level; and (iii) determining that the Kasai procedure is likely a failure.

(8) In some embodiments, the APP expression level is APP mRNA level or protein level. In some embodiments, step (i) comprises a reverse transcription polymerase chain reaction (RT-PCR) or an immunoassay. In some embodiments, the liver sample is a liver biopsy. In some embodiments, the method further includes after step (iii) a step of performing at least one additional diagnostic test

for BA to confirm the prognosis-if the additional test for BA indicates presence of BA, one may conclude the treatment has likely failed. In some embodiments, the method may further include after step (iii) a step of providing the infant with further treatment such as liver transplant.

(9) In a third aspect, the present invention provides a kit for diagnosing BA, assessing risk of developing BA, or prognosing unsuccessful Kasai procedure in an infant. The kit comprises a first container containing at least one reagent for detecting the mRNA or protein level of APP and a second container containing a standard control sample having an average level of APP expression in healthy liver tissue.

(10) In some embodiments, the APP expression level is APP mRNA level or APP protein level. In some embodiments, the first container contains one or more reagents for an RT-PCR. In some embodiments, the first container contains one or more reagents for an immunoassay. In some embodiments, one or more reagents contained in the first container include a set of primers for PCR. In some embodiments, one or more reagents contained in the first container include an antibody against APP. In some embodiments, the kit further includes user instructions for using the kit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1. Principal component analysis of control and BA liver.

(2) FIG. 2. Expression of APP in BA bile ducts. Sections of livers of biliary atresia (A) and normal controls (B; Hepatoblastoma) were immuno-fluorescent stained for APP (red) and CK19 (green; bile duct marker). Photos of liver sections of six BA patients (BA1 to BA6) and two HB patients (HB1 and HB2) were shown for comparison. Regions highlighted were enlarged and shown on the right.

DEFINITIONS

(3) As used herein, an “amyloid precursor protein (APP)” refers to a membrane protein encoded by a gene located on chromosome 21 (in human). A highly conserved protein found in many species, there are several known isoforms of human APP due to alternative splicing, ranging in length from 639 to 770 amino acids, with certain isoforms preferentially expressed in neurons. An integral membrane protein expressed in many tissues and concentrated in the synapses of neurons, APP has been implicated as a regulator of synapse formation, neural plasticity, and iron export, although its precise functions are yet to be fully understood. The term “amyloid precursor protein” or “APP” is used herein to broadly encompass any isoforms of the protein such as naturally occurring homologues or orthologues or mutants, including proteins or peptides derived from the full-length APP such as peptides generated by its proteolysis, for instance, beta amyloid (A β), a polypeptide containing about 37 to about 49 amino acid residues, whose amyloid fibrillar form is the primary component of amyloid plaques found in the brains of Alzheimer's disease patients. Preferably, an APP is specifically recognized by an antibody that specifically recognizes the full-length APP, e.g., a polypeptide encoded by any one of the polynucleotide sequences set forth in Table 3.

(4) Biliary atresia (BA) is a disease of blocked biliary system in the liver and bile ducts that typically occurs in infants in the age range of 2 weeks to 2 months. Symptoms of biliary atresia include jaundice, dark urine, acholic stools (clay-colored stools), weight loss and irritability. The preferred treatment method for BA is the Kasai procedure or operation, a surgical bypass procedure that allows bile to drain from the liver thus avoiding permanent damage to the liver. As used herein, a subject suffering from BA is an infant (for examples, aged about 1 week or about 2 weeks or about 3 weeks to about 1 month or about 2 months or about 3 months) who has been diagnosed with BA but is yet to receive the Kasai operation, or who has been diagnosed with BA, received the Kasai operation, yet still has symptoms relevant to the disease despite having received the Kasai

operation.

(5) In this disclosure the term “biological sample” or “sample” includes sections of tissues such as biopsy and autopsy samples, and frozen sections taken for histologic purposes, or processed forms of any of such samples. Biological samples include blood and blood fractions or products (e.g., serum, plasma, platelets, red blood cells, and the like), sputum or saliva, lymph and tongue tissue, oral swab, cultured cells, e.g., primary cultures, explants, transformed cells, stool, urine, and biopsy taken from a pre-selected organ or tissue (such as liver cells or hepatic tissue sample) as well as cells or tissue derived from biopsy. A biological sample is typically obtained from a eukaryotic organism, which may be a mammal, may be a primate and may be a human subject.

(6) The term “nucleic acid” or “polynucleotide” refers to deoxyribonucleotides or ribonucleotides and polymers thereof in either single- or double-stranded form. Unless specifically limited, the term encompasses nucleic acids containing known analogues of natural nucleotides which have similar binding properties as the reference nucleic acid and are metabolized in a manner similar to naturally occurring nucleotides. Unless otherwise indicated, a particular nucleic acid sequence also implicitly encompasses conservatively modified variants thereof (e.g., degenerate codon substitutions) and complementary sequences as well as the sequence explicitly indicated.

Specifically, degenerate codon substitutions may be achieved by generating sequences in which the third position of one or more selected (or all) codons is substituted with mixed-base and/or deoxyinosine residues (Batzer et al., *Nucleic Acid Res.*, 19:5081 (1991); Ohtsuka et al., *J. Biol. Chem.*, 260:2605-2608 (1985); and Cassol et al., (1992); Rossolini et al., *Mol. Cell. Probes*, 8:91-98 (1994)). The terms nucleic acid and polynucleotide are used interchangeably with gene, cDNA, and mRNA encoded by a gene.

(7) The terms “polypeptide,” “peptide,” and “protein” are used interchangeably herein to refer to a polymer of amino acid residues. The terms apply to amino acid polymers in which one or more amino acid residue is an artificial chemical mimetic of a corresponding naturally occurring amino acid, as well as to naturally occurring amino acid polymers and non-naturally occurring amino acid polymers. As used herein, the terms encompass amino acid chains of any length, including full length proteins (i.e., antigens), wherein the amino acid residues are linked by covalent peptide bonds.

(8) The term “amino acid” refers to naturally occurring and synthetic amino acids, as well as amino acid analogs and amino acid mimetics that function in a manner similar to the naturally occurring amino acids. Naturally occurring amino acids are those encoded by the genetic code, as well as those amino acids that are later modified, e.g., hydroxyproline, γ -carboxyglutamate, and O-phosphoserine. Amino acid analogs refers to compounds that have the same basic chemical structure as a naturally occurring amino acid, i.e., an α carbon that is bound to a hydrogen, a carboxyl group, an amino group, and an R group, e.g., homoserine, norleucine, methionine sulfoxide, methionine methyl sulfonium. Such analogs have modified R groups (e.g., norleucine) or modified peptide backbones, but retain the same basic chemical structure as a naturally occurring amino acid. “Amino acid mimetics” refers to chemical compounds that have a structure that is different from the general chemical structure of an amino acid, but that functions in a manner similar to a naturally occurring amino acid.

(9) Amino acids may be referred to herein by either their commonly known three letter symbols or by the one-letter symbols recommended by the IUPAC-IUB Biochemical Nomenclature Commission. Nucleotides, likewise, may be referred to by their commonly accepted single-letter codes.

(10) A “polynucleotide hybridization method” as used herein refers to a method for detecting the presence and/or quantity of a pre-determined polynucleotide sequence based on its ability to form Watson-Crick base-pairing, under appropriate hybridization conditions, with a polynucleotide probe of a known sequence. Examples of such hybridization methods include Southern blot, Northern blot, and in situ hybridization.

(11) “Primers” as used herein refer to oligonucleotides that can be used in an amplification method, such as a polymerase chain reaction (PCR), to amplify a nucleotide sequence based on the polynucleotide sequence corresponding to a gene of interest, e.g., the polynucleotide sequence encoding an APP or a derivative thereof. Typically at least one, possibly two, of the PCR primers for amplification of a polynucleotide sequence is sequence-specific for that polynucleotide sequence. The exact length of the primer will depend upon many factors, including temperature, source of the primer, and the method used. For example, for diagnostic and prognostic applications, depending on the complexity of the target sequence, the oligonucleotide primer typically contains at least 10, or 15, or 20, or 25 or more nucleotides, although it may contain fewer nucleotides or more nucleotides. The factors involved in determining the appropriate length of primer are readily known to one of ordinary skill in the art. The primers used in particular embodiments may be designed to specifically amplify a coding sequence for an APP or its variant or derivative. Also, the primers may be designed for specifically amplify only one segment of the APP coding sequence for a fragment of APP. In this disclosure the term “primer pair” means a pair of primers that hybridize to opposite strands a target DNA molecule or to regions of the target DNA which flank a nucleotide sequence to be amplified (e.g., encoding an APP for the purpose of detection such as for assessing BA risk or disease prospect, especially after Kasai operation). In this disclosure the term “primer site” means the area of the target DNA or other nucleic acid to which a primer hybridizes.

(12) A “label,” “detectable label,” or “detectable moiety” is a composition detectable by spectroscopic, photochemical, biochemical, immunochemical, chemical, or other physical means. For example, useful labels include radioactive isotopes, fluorescent dyes, electron-dense reagents, enzymes (e.g., as commonly used in an ELISA), biotin, digoxigenin, or haptens and proteins that can be made detectable, e.g., by incorporating a radioactive component into the peptide or used to detect antibodies specifically reactive with the peptide. Typically a detectable label is attached to a probe or a molecule with defined binding characteristics (e.g., a polypeptide with a known binding specificity or a polynucleotide), so as to allow the presence of the probe (and therefore its binding target, e.g., a coding sequence for an APP or a fragment or variant thereof) to be readily detectable.

(13) The term “treat” or “treating,” as used in this application, describes to an act that leads to the elimination, reduction, alleviation, reversal, or prevention or delay of onset or recurrence of any symptom of a relevant condition (e.g., BA). In other words, “treating” a condition encompasses both therapeutic and prophylactic intervention against the condition: for example, upon testing positive to have elevated APP expression either at mRNA or protein level, an infant will be given additional diagnostic tests (such as blood tests, ultrasound of the abdomen, liver biopsy, as well as diagnostic surgery, which may include an operative cholangiogram) and, upon confirmation of the diagnosis of BA, undergoes therapeutic or prophylactic regimen under the supervision of a medical professional including Kasai operation. For an infant who has received treatment for BA by Kasai operation but still has BA symptoms without improvement or even with worsening symptoms, subsequent treatment will likely require liver transplant depending on physicians' decision upon careful monitoring of the infant. Administration of steroids and/or antibiotics may serve as additional/adjuvant treatment post-Kasai operation with an aim to slow down or halt further inflammation.

(14) The term “effective amount” as used herein refers to an amount of a given substance that is sufficient in quantity to produce a desired effect. For example, when an effective amount of a therapeutic agent for treating BA by way of suppressing or eliminating the expression and/or activity of APP especially in an infant's liver is administered to the infant, the symptoms of BA are reduced, reversed, eliminated, prevented, or delayed of the onset in the infant. An amount adequate to accomplish this is defined as the “therapeutically effective dose” when administration takes place after BA symptoms have become detectable, or is defined as the “prophylactically effective dose” when administration takes place before any symptom has arisen. The dosing range varies with the nature of the therapeutic agent being administered and other factors such as the route of

administration and the severity of a patient's BA and related condition.

(15) The term "subject" or "subject in need of treatment," as used herein, includes individuals who seek medical attention due to risk of, or actual suffering from, biliary atresia. Subjects also include individuals currently undergoing therapy that seek manipulation of the therapeutic regimen.

Subjects or individuals in need of treatment include those that demonstrate symptoms of biliary atresia, or those that are at risk of suffering from biliary atresia or its symptoms, or those that have been tested positive for amyloid precursor protein expression but may or may not have any known risk and/or may or may not have any potentially relevant symptoms. For example, subjects in need of treatment include individuals with a genetic predisposition or family history for biliary atresia, those that are acutely suffering from relevant symptoms, those that have been exposed to a triggering substance or event, as well as those that have received treatment for BA (e.g., Kasai operation) but are continuing to suffer from symptoms of the condition. A subject in need of treatment for BA may be any gender and typically an infant at the age of 3 months or younger, for example, between about 1 week to about 3 months, or between about 2 weeks to about 2 months.

(16) The term "about" when used in reference to a pre-determined value denotes a range encompassing $\pm 10\%$ of the value.

DETAILED DESCRIPTION

I. General

(17) The present invention relates to the detection of APP expression in the liver cells or tissue of infants who are suffering from or at risk of suffering from biliary atresia, or who are at risk of suffering from undesirable clinical outcome after receiving the Kasai operation. This invention thus provides methods for diagnosis of biliary atresia and prognosis for clinical outcome post-Kasai operation in infants based on detection of APP expression in a liver sample obtained from infants suspected of suffering from biliary atresia or having undergone the Kasai procedure.

(18) Basic texts disclosing general methods and techniques in the field of recombinant genetics include Sambrook and Russell, *Molecular Cloning, A Laboratory Manual* (3rd ed. 2001); Kriegler, *Gene Transfer and Expression: A Laboratory Manual* (1990); and Ausubel et al., eds., *Current Protocols in Molecular Biology* (1994).

(19) For nucleic acids, sizes are given in either kilobases (kb) or base pairs (bp). These are estimates derived from agarose or acrylamide gel electrophoresis, from sequenced nucleic acids, or from published DNA sequences. For proteins, sizes are given in kilodaltons (kDa) or amino acid residue numbers. Protein sizes are estimated from gel electrophoresis, from sequenced proteins, from derived amino acid sequences, or from published protein sequences.

(20) Oligonucleotides that are not commercially available can be chemically synthesized, e.g., according to the solid phase phosphoramidite triester method first described by Beaucage & Caruthers, *Tetrahedron Lett.* 22:1859-1862 (1981), using an automated synthesizer, as described in Van Devanter et al., *Nucleic Acids Res.* 12:6159-6168 (1984). Purification of oligonucleotides is performed using any art-recognized strategy, e.g., native acrylamide gel electrophoresis or anion-exchange HPLC as described in Pearson & Reanier, *J. Chrom.* 255:137-149 (1983).

(21) The polynucleotide sequence encoding a polypeptide of interest, e.g., an amyloid precursor protein or a variant or derivative thereof, and synthetic oligonucleotides can be verified after cloning or subcloning using, e.g., the chain termination method for sequencing double-stranded templates of Wallace et al., *Gene* 16:21-26 (1981).

II. Acquisition of Samples and Analysis of mRNA

(22) The present invention relates to detecting the expression level of APP, either at the mRNA or protein level, in the liver cells or tissue taken from an infant suspected of suffering from biliary atresia (BA) or an infant who has been diagnosed with BA and has received the Kasai operation, as a means to detect the presence of BA, to assess the risk of developing of BA, and/or to provide prognosis of BA treatment such as the Kasai operation in an infant, so as to allow the correct treatment and/or follow-up strategies after Kasai operation. Thus, the first steps of practicing this

invention are to obtain an appropriate sample from a test subject and extract mRNA from the sample.

(23) A. Acquisition and Preparation of Samples

(24) An appropriate sample such as a liver cell or tissue sample obtained from an infant, who may or may not exhibit BA symptoms, to be tested or monitored for BA or treatment outcome using a method of the present invention. Collection of a liver biopsy sample from an individual is performed in accordance with the standard protocol hospitals or clinics generally follow. An appropriate amount of tissue sample, including biopsy samples, is collected and may be stored according to standard procedures prior to further preparation.

(25) B. Extraction of RNA

(26) There are numerous methods for extracting RNA from a biological sample. The general methods of RNA preparation (e.g., described by Sambrook and Russell, *Molecular Cloning: A Laboratory Manual* 3d ed., 2001) can be followed; various commercially available reagents or kits, such as Trizol reagent (Invitrogen, Carlsbad, CA), Oligotex Direct mRNA Kits (Qiagen, Valencia, CA), RNeasy Mini Kits (Qiagen, Hilden, Germany), and PolyATtract® Series 9600™ (Promega, Madison, WI), may also be used to obtain RNA from a liver sample from an infant. Combinations of more than one of these methods may also be used.

(27) It is preferable in some applications that all or most of the contaminating DNA be eliminated from the RNA preparations. Thus, careful handling of the samples, thorough treatment with DNase, and proper negative controls in the amplification and quantification steps should be used.

(28) C. PCR-Based Quantitative Determination of RNA Level

(29) Once RNA is extracted from a biological sample, the amount of RNA derived from a genetic locus of interest, e.g., and encoding for a protein of interest such as APP, may be quantified. The preferred method for determining the RNA level is an amplification-based method, e.g., by PCR.

(30) Prior to the amplification step, a DNA copy (cDNA) of the RNA of interest must be synthesized. This is achieved by reverse transcription, which can be carried out as a separate step, or in a homogeneous reverse transcription-polymerase chain reaction (RT-PCR), a modification of the polymerase chain reaction for amplifying RNA. Methods suitable for PCR amplification of ribonucleic acids are described by Romero and Rotbart in *Diagnostic Molecular Biology: Principles and Applications* pp. 401-406; Persing et al., eds., Mayo Foundation, Rochester, MN, 1993; Egger et al., *J. Clin. Microbiol.* 33:1442-1447, 1995; and U.S. Pat. No. 5,075,212.

(31) The general methods of PCR are well known in the art and are thus not described in detail herein. For a review of PCR methods, protocols, and principles in designing primers, see, e.g., Innis, et al., *PCR Protocols: A Guide to Methods and Applications*, Academic Press, Inc. N.Y., 1990. PCR reagents and protocols are also available from commercial vendors, such as Roche Molecular Systems.

(32) PCR is most usually carried out as an automated process with a thermostable enzyme. In this process, the temperature of the reaction mixture is typically cycled through a denaturing region, a primer annealing region, and an extension reaction region automatically. In some protocols, the annealing region and the extension reaction region are merged. Machines specifically adapted for this purpose are commercially available.

(33) Although PCR amplification of the target RNA is typically used in practicing the present invention. One of skill in the art will recognize, however, that amplification of these RNA species in a maternal blood sample may be accomplished by any known method, such as ligase chain reaction (LCR), transcription-mediated amplification, and self-sustained sequence replication or nucleic acid sequence-based amplification (NASBA), each of which provides sufficient amplification. More recently developed branched-DNA technology may also be used to quantitatively determining the amount of RNA markers in maternal blood. For a review of branched-DNA signal amplification for direct quantitation of nucleic acid sequences in clinical samples, see Nolte, *Adv. Clin. Chem.* 33:201-235, 1998.

(34) C. Other Quantitative Methods

(35) The RNA species of interest (such as the APP mRNA) can also be detected using other standard techniques, well-known to those of skill in the art. Although the detection step is typically preceded by an amplification step, amplification is not required in the methods of the invention. For instance, the RNA species of interest may be identified by size fractionation (e.g., gel electrophoresis), whether or not preceded by an amplification step. After running a sample in an agarose or polyacrylamide gel and labeling with ethidium bromide according to well-known techniques (see, e.g., Sambrook and Russell, *supra*), the presence of a band of the same size as the standard control is an indication of the presence of a target RNA, the amount of which may then be compared to the control based on the intensity of the band. Alternatively, oligonucleotide probes specific to RNA transcribed from a genetic locus, e.g., the APP gene, can be used to detect the presence of such RNA species and indicate the amount of RNA in comparison to the standard control, based on the intensity of signal imparted by the probe.

(36) Sequence-specific probe hybridization is a well-known method of detecting a particular nucleic acid comprising other species of nucleic acids. Under sufficiently stringent hybridization conditions, the probes hybridize specifically only to substantially complementary sequences. The stringency of the hybridization conditions can be relaxed to tolerate varying amounts of sequence mismatch.

(37) A number of hybridization formats are well-known in the art, including but not limited to, solution phase, solid phase, or mixed phase hybridization assays. The following articles provide an overview of the various hybridization assay formats: Singer et al., *Biotechniques* 4:230, 1986; Haase et al., *Methods in Virology*, pp. 189-226, 1984; Wilkinson, *In situ Hybridization*, Wilkinson ed., IRL Press, Oxford University Press, Oxford; and Hames and Higgins eds., *Nucleic Acid Hybridization: A Practical Approach*, IRL Press, 1987.

(38) The hybridization complexes are detected according to well-known techniques and the detection is not a critical aspect of the present invention. Nucleic acid probes capable of specifically hybridizing to a target nucleic acid, i.e., the APP mRNA species or the amplified DNA, can be labeled by any one of several methods typically used to detect the presence of hybridized nucleic acids. One common method of detection is the use of autoradiography using probes labeled with .sup.3H, .sup.125I, .sup.35S, .sup.14C, or .sup.32P, or the like. The choice of radioactive isotope depends on research preferences due to ease of synthesis, stability, and half-lives of the selected isotopes. Other labels include compounds (e.g., biotin and digoxigenin), which bind to antigens or antibodies labeled with fluorophores, chemiluminescent agents, and enzymes.

Alternatively, probes can be conjugated directly with labels such as fluorophores, chemiluminescent agents or enzymes. The choice of label depends on sensitivity required, ease of conjugation with the probe, stability requirements, and available instrumentation.

(39) The probes and primers necessary for practicing the present invention can be synthesized and labeled using well known techniques. Oligonucleotides used as probes and primers may be chemically synthesized according to the solid phase phosphoramidite triester method first described by Beaucage and Caruthers, *Tetrahedron Letts.*, 22:1859-1862, 1981, using an automated synthesizer, as described in Needham-VanDevanter et al., *Nucleic Acids Res.* 12:6159-6168, 1984. Purification of oligonucleotides is by either native acrylamide gel electrophoresis or by anion-exchange high performance liquid chromatography (HPLC) as described in Pearson and Regnier, *J. Chrom.*, 255:137-149, 1983.

III. Immunoassays for Detection of Amyloid Precursor Protein

(40) One aspect of this invention provides immunoassays used in the detection of APP in order to determine the expression level of the protein for the purpose of diagnosis and prognosis of BA. Antibodies against APP described herein are useful for carrying out these immunological assays.

(41) A. Production of Antibodies Against APP

(42) Methods for producing polyclonal and monoclonal antibodies that react specifically with an

immunogen of interest are known to those of skill in the art (see, e.g., Coligan, *Current Protocols in Immunology* Wiley/Greene, NY, 1991; Harlow and Lane, *Antibodies: A Laboratory Manual* Cold Spring Harbor Press, NY, 1989; Stites et al. (eds.) *Basic and Clinical Immunology* (4th ed.) Lange Medical Publications, Los Altos, CA, and references cited therein; Goding, *Monoclonal Antibodies: Principles and Practice* (2d ed.) Academic Press, New York, NY, 1986; and Kohler and Milstein *Nature* 256:495-497, 1975). Such techniques include antibody preparation by selection of antibodies from libraries of recombinant antibodies in phage or similar vectors (see, Huse et al., *Science* 246:1275-1281, 1989; and Ward et al., *Nature* 341:544-546, 1989).

(43) In order to produce antisera containing antibodies with desired specificity, the polypeptide of interest (e.g., APP) or an antigenic fragment thereof can be used to immunize suitable animals, e.g., mice, rabbits, or primates. A standard adjuvant, such as Freund's adjuvant, can be used in accordance with a standard immunization protocol. Alternatively, a synthetic antigenic peptide derived from that particular polypeptide can be conjugated to a carrier protein and subsequently used as an immunogen.

(44) The animal's immune response to the immunogen preparation is monitored by taking test bleeds and determining the titer of reactivity to the antigen of interest. When appropriately high titers of antibody to the antigen are obtained, blood is collected from the animal and antisera are prepared. Further fractionation of the antisera to enrich antibodies specifically reactive to the antigen and purification of the antibodies can be performed subsequently, see, Harlow and Lane, *supra*, and the general descriptions of protein purification provided above.

(45) Monoclonal antibodies are obtained using various techniques familiar to those of skill in the art. Typically, spleen cells from an animal immunized with a desired antigen are immortalized, commonly by fusion with a myeloma cell (see, Kohler and Milstein, *Eur. J. Immunol.* 6:511-519, 1976). Alternative methods of immortalization include, e.g., transformation with Epstein Barr Virus, oncogenes, or retroviruses, or other methods well known in the art. Colonies arising from single immortalized cells are screened for production of antibodies of the desired specificity and affinity for the antigen, and the yield of the monoclonal antibodies produced by such cells may be enhanced by various techniques, including injection into the peritoneal cavity of a vertebrate host.

(46) Additionally, monoclonal antibodies may also be recombinantly produced upon identification of nucleic acid sequences encoding an antibody with desired specificity or a binding fragment of such antibody by screening a human B cell cDNA library according to the general protocol outlined by Huse et al., *supra*. The general principles and methods of recombinant polypeptide production discussed above are applicable for antibody production by recombinant methods.

(47) B. Immunoassays for APP

(48) Once antibodies specific for APP are available, the expression of APP at protein level in a sample, e.g., a liver cell or tissue sample, can be measured by a variety of immunoassay methods providing qualitative and quantitative results to a skilled artisan. For a review of immunological and immunoassay procedures in general see, e.g., Stites, *supra*; U.S. Pat. Nos. 4,366,241; 4,376,110; 4,517,288; and 4,837,168.

(49) 1. Labeling in Immunoassays

(50) Immunoassays often utilize a labeling agent to specifically bind to and label the binding complex formed by the antibody and the target protein (e.g., APP). The labeling agent may itself be one of the moieties comprising the antibody/target protein complex, or may be a third moiety, such as another antibody, that specifically binds to the antibody/target protein complex. A label may be detectable by spectroscopic, photochemical, biochemical, immunochemical, electrical, optical or chemical means. Examples include, but are not limited to, magnetic beads (e.g., Dynabeads™), fluorescent dyes (e.g., fluorescein isothiocyanate, Texas red, rhodamine, and the like), radiolabels (e.g., .sup.3H, .sup.125I, .sup.35S, .sup.14C, or .sup.32P), enzymes (e.g., horse radish peroxidase, alkaline phosphatase, and others commonly used in an ELISA), and colorimetric labels such as colloidal gold or colored glass or plastic (e.g., polystyrene, polypropylene, latex, etc.) beads.

(51) In some cases, the labeling agent is a second antibody bearing a detectable label. Alternatively, the second antibody may lack a label, but it may, in turn, be bound by a labeled third antibody specific to antibodies of the species from which the second antibody is derived. The second antibody can be modified with a detectable moiety, such as biotin, to which a third labeled molecule can specifically bind, such as enzyme-labeled streptavidin.

(52) Other proteins capable of specifically binding immunoglobulin constant regions, such as protein A or protein G, can also be used as the label agents. These proteins are normal constituents of the cell walls of streptococcal bacteria. They exhibit a strong non-immunogenic reactivity with immunoglobulin constant regions from a variety of species (see, generally, Kronval, et al. *J. Immunol.*, 111:1401-1406 (1973); and Akerstrom, et al., *J. Immunol.*, 135:2589-2542 (1985)).

(53) 2. Immunoassay Formats

(54) Immunoassays for detecting a target protein of interest (e.g., APP) from samples may be either competitive or noncompetitive. Noncompetitive immunoassays are assays in which the amount of captured target protein is directly measured. In one preferred "sandwich" assay, for example, the antibody specific for the target protein can be bound directly to a solid substrate where the antibody is immobilized. It then captures the target protein in test samples. The antibody/target protein complex thus immobilized is then bound by a labeling agent, such as a second or third antibody bearing a label, as described above.

(55) In competitive assays, the amount of target protein in a sample is measured indirectly by measuring the amount of an added (exogenous) target protein displaced (or competed away) from an antibody specific for the target protein by the target protein present in the sample. In a typical example of such an assay, the antibody is immobilized and the exogenous target protein is labeled. Since the amount of the exogenous target protein bound to the antibody is inversely proportional to the concentration of the target protein present in the sample, the target protein level in the sample can thus be determined based on the amount of exogenous target protein bound to the antibody and thus immobilized.

(56) In some cases, western blot (immunoblot) analysis is used to detect and quantify the presence of APP in the samples. The technique generally comprises separating sample proteins by gel electrophoresis on the basis of molecular weight, transferring the separated proteins to a suitable solid support (such as a nitrocellulose filter, a nylon filter, or a derivatized nylon filter) and incubating the samples with the antibodies that specifically bind the target protein. These antibodies may be directly labeled or alternatively may be subsequently detected using labeled antibodies (e.g., labeled sheep anti-mouse antibodies) that specifically bind to the antibodies against APP.

(57) Other assay formats include liposome immunoassays (LIA), which use liposomes designed to bind specific molecules (e.g., antibodies) and release encapsulated reagents or markers. The released chemicals are then detected according to standard techniques (see, Monroe et al., *Amer. Clin. Prod. Rev.*, 5:34-41 (1986)).

(58) For these immunoassays, the patient being tested may be an infant who is at risk of developing BA or may be one who has been diagnosed with BA, has undergone treatment such as Kasai operation, and is now being assessed for likelihood of a desirable outcome of the treatment.

IV. Treatment Options

(59) One practical application of this invention is intended for early detection of BA among infants who are suspected of suffering from the disease or are at increased risk of developing the disease. Once they are tested according to any one of the methods of this invention and are determined as at risk of suffering from or later developing this disease, they can be subject to additional diagnostic tests, such as blood tests, ultrasound of the abdomen, liver biopsy, and diagnostic surgery (which may include an operative cholangiogram) in order to confirm whether they indeed suffer from the disease. Upon confirmation of the diagnosis of BA, they can then undergo appropriate treatment (e.g., the Kasai operation) as prescribed by their attending physician.

(60) Another application is intended for predicting the likelihood of success among infants who have been diagnosed with BA and have been given appropriate treatment such as the Kasai operation. As it is known that a minority of all infant BA patients who underwent the operation will fail to properly recover, and the symptoms and damaging effects of BA will continue and even worsen. The claimed method of this invention allows physicians to quickly obtain an assessment of the likelihood of treatment outcome (e.g., shortly after the Kasai procedure such as 1 day, 2 days, 3 days, 4 days, 5 days, 6 days, 7 days, or up to 10 days or 2 weeks post-operation) when the infant may or may not yet to show any signs of persistent or deteriorating BA symptoms: whether it will result in desirable outcome, where the infant patient will fully recover from the disease, or it will result in failure, where the infant patient will require additional treatment, such as another Kasai procedure or even liver transplant.

(61) A further aspect of the present invention is the treatment of BA. Since the expression of APP, both at mRNA level and at protein level, has been found to increase in the liver of infants who suffer from BA, or who are at increased risk of later developing BA while exhibiting no BA symptoms for the time being, or who have been diagnosed with BA and given appropriate treatment such as the Kasai procedure but suffer from undesirable outcome (i.e., failure of treatment to alleviate BA symptoms and damages to liver), one treatment method for BA is to target APP by suppressing its expression (both at mRNA and protein level) and/or activity. For instance, specific inhibitors targeting APP expression and/or activity may be administered to an infant who suffers from BA, or who is at heightened risk of later developing BA, or who has received a diagnosis of BA and received treatment (e.g., the Kasai procedure) but is at risk of suffering from treatment failure. Such specific inhibitors include a broad spectrum of possible compounds of distinct chemical and structural features such as a dominant negative APP mutant or its encoding nucleic acid, a nucleic acid encoding an antisense or miRNA, miniRNA, long non-coding RNA targeting APP, an inactivating anti-APP antibody, small chemicals, peptides, proteins, natural extract compounds from herbs, etc., and they are useful in both prophylactic and therapeutic applications for treating BA. Also, APP expression may be suppressed by genetic manipulation techniques including CRISPR.

V. Establishing Standard Controls

(62) For the application of this invention for the purpose of diagnosing BA or assessing risk of developing BA at a later time, in order to establish a standard control, a group of healthy infants without any liver disorders, especially BA, should first be selected before liver samples are obtained. These infants should be of similar age, which is within the appropriate time frame (for example, aged from about 2 weeks to about 8 weeks) when infants may be tested using the methods of the present invention. The health status of the selected infants should be confirmed by well established, routinely employed methods including but not limited to blood tests and abdominal ultrasound.

(63) The selected group of healthy infants must be of a reasonable size, such that the average amount of APP mRNA and protein found in the liver tissue samples (e.g., liver biopsy) calculated from the group can be reasonably regarded as representative of the normal or average amount among the general population of healthy infants not suffering from BA and not at heightened risk of developing BA. Preferably, the selected group comprises at least 10 healthy infants.

(64) For the application of this invention for the purpose of providing a prognosis of BA treatment (especially the Kasai procedure) outcome among infants who have already been diagnosed with BA and received appropriate treatment such as the Kasai procedure, the same standard control described above may be used. In the alternative, a slightly different standard control can be established: a group of infants who suffered from BA and have undergone the same BA treatment such as Kasai operation and successfully recovered should be selected for collection of liver samples. Similar to the description above, these infants should be of similar age and make up a group of a reasonable size, e.g., at least 10 in the group.

(65) Once an average value is established for the amount of APP mRNA or protein based on the individual values found in the liver tissue of each infant of the selected group, this value is considered a standard control value for APP mRNA or protein. Any liver tissue sample that contains a similar amount of APP mRNA or protein can thus be used as a standard control sample.

VI. Kits

(66) The invention also provides kits for detecting APP mRNA or protein, especially in a liver sample taken from an infant, according to the method of the present invention for the purpose of detecting the presence of BA, assessing the risk of later developing BA, or assessing likelihood of a successful BA treatment (such as the Kasai operation) outcome. The kits typically include a first container that contains a first container containing at least one, possibly more, reagent for detecting expression level of APP and a second container containing a composition having a standard control level of APP expression (or a standard control sample). The one or more reagents in the first container may be used for measuring APP mRNA level, such as oligonucleotide primers useful for RT-PCR and/or oligonucleotide probe(s) for specific hybridization with an APP-specific polynucleotide sequence. Other possible reagents may include the necessary enzyme(s) and buffer(s) for performing assays to detect APP mRNA such as RT-PCR or nucleotide hybridization.

(67) For the purpose of determining the APP protein level in a liver sample, the one or more reagents contained in the first container may be those useful for immunoassays capable of specifically detecting and quantifying APP level as a protein, such as ELISA or western blot analysis. One useful reagent is an anti-APP antibody, which may be a polyclonal antibody or a monoclonal antibody capable of specific binding to APP. Further included may be a secondary antibody, for example, an antibody against the anti-APP antibody, preferably conjugated with a detectable label.

(68) In addition, the kit may further include informational material containing instructions for a user on how to use the kit for performing an assay and determining whether increased APP mRNA or protein is present in a patient sample.

EXAMPLES

(69) The following examples are provided by way of illustration only and not by way of limitation. Those of skill in the art will readily recognize a variety of non-critical parameters that could be changed or modified to yield essentially the same or similar results.

Example 1: Amyloid Precursor Protein Expression in Biliary Atresia

(70) This study reveals, for the first time, the specific expression of amyloid precursor protein and/or its processed forms in the bile duct of livers in patients suffering from biliary atresia (BA). This discovery allows one to devise new and reliable diagnostic methods for early detection of BA as well as for predicting the therapeutic outcome of BA treatment such as Kasai operation.

(71) Introduction

(72) Biliary atresia (BA [OMIM 210500]) is characterized by progressive fibro-obliterative cholangiopathy (disease of the bile duct) affecting both the intra- and extra-hepatic bile ducts and resulting in obstructive bile flow, cholestasis, and jaundice in neonates. BA occurs in some infants two to six weeks after birth, and symptoms of BA are usually evident between two and six weeks after birth, in that babies with BA develop progressive cholestasis, a condition in which the liver is unable to excrete bilirubin through the bile ducts in the form of bile. Bilirubin builds up inside of liver and begins to accumulate in the blood, causing symptoms including yellowing of the skin, itchiness, poor absorption of nutrients, pale stools, dark urine, and a swollen abdomen.

(73) The differential diagnosis of BA allows for identification of patients suffering from BA as opposed to other disorders such as neonatal cholestasis (NC) like neonatal hepatitis (NH), paucity of interlobular bile ducts (PILBD), progressive familial intrahepatic cholestasis (PFIC), and various metabolic diseases like galactosemia and α -1 antitrypsin deficiency. The most important objective in such cases is to distinguish obstructive cholestasis from non-obstructive causes. The diagnosis of BA, particularly distinguishing it from other causes of liver injury in the neonatal period, is

challenging as there is a high degree of overlap in clinical, biochemical, imaging, and histological characteristics of BA and other causes of NC.

(74) Without surgical treatment (Kasai operation) to re-establish the bile drainage from microscopic residual bile ductules within the liver, progressive hepatic fibrosis leads to cirrhosis, portal vein hypertension, liver failure and death by the age of two (1, 2). However, postsurgical complications, including cholangitis (50%) and portal hypertension (>60%), remain a problem (3-7). Furthermore, regardless of drainage after successful surgery, patients will often develop inflammation and sclerosis of the biliary tree, leading to secondary biliary cirrhosis. To these patients and those who failed the surgical intervention, liver transplantation becomes the only treatment option. Indeed, BA has been the most common indication for liver transplantation during childhood for the past 20 years. Due to liver graft scarcity, many patients die before transplantation.

(75) Urgently, there exists a distinct need to address the following problems: first, diagnosis of BA needs to be improved, in particular at early stages of the disease. The optimal age for Kasai operation is 60-75 days; delayed in diagnosis and surgery will lead to poor outcome. In fact, early diagnosis allows the family and surgeon for better planning of disease management. Second, despite the same Kasai operation, the outcome can vary from patient to patient, suggesting the existence of subsets of patient but there are no reliable markers for prognosis to guide stratified treatments for optimal outcomes. Third, regardless of drainage after Kasai operation, patients will often develop bile duct inflammation and sclerosis, leading to secondary biliary cirrhosis. To these patients and those who failed surgery, liver transplantation is the only option. For those BA patients who are predicted to have poor outcome after Kasai operation may be selected to undergo liver transplantation as the treatment from the start without first undergoing Kasai operation. There is an urgent need to find new and effective therapies that ameliorate BA.

(76) Materials and Methods

(77) Liver Tissues

(78) Control liver: “Quasi-normal” human liver tissues were obtained from non-tumour margin of hepatoblastoma (HB).

(79) BA liver: Liver biopsies of BA patients taken during Kasai operation or at transplantation. All tissues were obtained during operations with full informed consent from parents or patients, and the study was approved by Hong Kong West Cluster-Hong Kong University Cluster Research Ethics Committee/Institutional Review Board (UW 16-052).

Antibodies

(80) Anti- β -Amyloid (4G8; Biolegend) reacts to precursor forms, as well as abnormally processed isoforms of amyloid precursor protein. Anti-CK19 (ab52625; abcam) reacts to human cytokeratin 19.

(81) Liver Biopsies

(82) Wedge biopsies (50×50 mm) were obtained from non-syndromic BA patients during laparoscopic cholangiography, hepatoblastoma (HB)).

(83) RNA Sequencing of Liver Tissues

(84) Total RNAs were prepared from 30 mg of control livers (HB; n=2) and BA livers (n=4) using RNeasy Mini Kit (Qiagen) following manufacturer's protocol. Reverse transcription, amplification of 50 ng total RNA of each liver samples, and library construction were performed using single cell RNA-seq technology (Smart-seq 2.0) with minor modifications (8). Qualities of the pre-amplified products of normal and BA livers were confirmed to be optimal by Bioanalyzer. Library construction was performed using Nextera XT Kit following manufacturer's protocol. Libraries were pooled and sequenced by pair ends of 100 base pairs (PE100) on illumina HiSeq 2500 System.

(85) Immuno-Fluorescence Staining

(86) Tissues were fixed in 4% paraformaldehyde (w/v) in PBS (phosphate-buffered saline, pH 7.2) for 48 h at 4° C., dehydrated in graded series of alcohol, cleared in xylene before being embedded

in paraffin. Sections (8 μ m in thickness) were prepared, mounted onto TESPA-coated microscope glass. Sections were dewaxed in xylene, hydrated in a graded series of alcohol and finally in distilled water. Antigen was retrieved by two steps: (i) incubating in 70% Formic acid for 10 minutes and washing in water; (ii) incubating in 10 mM sodium citrate buffer (pH 6.0) at 95° C. for 10 min and washing in water. After blocking in PBS-T (PBS with 0.1% Triton) supplemented with 1% Bovine Serum Albumin for 1 h at room temperature, sections were incubated with anti- β -Amyloid (4G8, 1:200; Biolegend) and anti-CK19 (ab52625, 1:250; abcam) diluted in PBS-T/BSA for overnight at 4° C. After washing in PBS-T, sections were incubated with appropriate fluorescent tagged secondary antibodies in PBS-T/BSA at 37° C. for 1 h. After PBS-T washings, sections were mounted in Dapi-containing anti-fade mounting fluid. Images were taken with Nikon Eclipse 80i microscope mounted with SPOT RT3 microscope digital camera under fluorescence illumination. Photos were compiled using Adobe Photoshop 7.

(87) Results

(88) Transcriptome Analysis of Normal and BA Liver by RNAseq

(89) The RNA-seq reads were first subjected to quality check using FastQC version 0.11.1. Further the adapter contamination and low-quality regions were filtered using Cutadapt version 1.8.3 with the parameter $-q=33$ and retained only reads with $\text{length} \geq 30$. The percentage of high quality bases in the filtered raw reads were greater than 90% for all the samples, with the quality score cut-off=20. Subsequently, sequencing reads were filtered for rRNA sequence by aligning to human rRNA sequences using Bowtie 2 (default parameters). The remaining reads were mapped to the reference genome. The transcriptome mapping/alignment and identification of exon-exon splice junctions with the human genome reference (GRCh38, downloaded from Ensembl database) was done by using TopHat version 2.0.10 (default parameters). All the samples had an overall alignment of >80% with human reference. Counting of aligned reads per gene were done using HTSeq version 0.9.1 for further differential expression analysis. The counts for each gene per samples were presented as table in DESeq2 to accurately detect significant differentially expressed genes across the conditions. Visualization was done using R and Bioconductor (FIG. 5c?).

(90) Principal component analysis (PCA) of liver biopsies revealed the following observations: 1. Normal liver and BA liver displayed distinctive molecular signatures. 2. The two normal liver controls showed differences among themselves, which may indicate the genetic differences between the patients or the tissue stages during isolation. 3. Three of the BA livers were clustering very closely to each other while the other patient showed enormous difference, which may indicate progression of disease.

(91) TABLE-US-00001 TABLE 1 Pathways involvement of the 1558 differentially expressed genes
Integrin signalling pathway (P00034) 28 Gonadotropin-releasing hormone receptor pathway (P06664) 25 Inflammation mediated by chemokine and cytokine signaling pathway (P00031) 21 CCKR signaling map (P06959) 20 Wnt signaling pathway (P00057) 17 Angiogenesis (P00005) 12 PDGF signaling pathway (P00047) 12 Apoptosis signaling pathway (P00006) 11 Nicotinic acetylcholine receptor signaling pathway (P00044) 10 Huntington disease (P00029) 9 Alzheimer disease-presenilin pathway (P00004) 8 Heterotrimeric G-protein signaling pathway-Gq alpha and Go alpha mediated pathway 8 (P00027) Heterotrimeric G-protein signaling pathway-Gi alpha and Gs alpha mediated pathway 8 (P00026) Parkinson disease (P00049) 7 Interleukin signaling pathway (P00036) 7 EGF receptor signaling pathway (P00018) 7 T cell activation (P00053) 6 TGF-beta signaling pathway (P00052) 6 FGF signaling pathway (P00021) 6 FAS signaling pathway (P00020) 6 Cadherin signaling pathway (P00012) 6 B cell activation (P00010) 6 Axon guidance mediated by Slit/Robo (P00008) 5 Toll receptor signaling pathway (P00054) 5 Oxidative stress response (P00046) 5 Muscarinic acetylcholine receptor 1 and 3 signaling pathway (P00042) 5 Insulin/IGF pathway-protein kinase B signaling cascade (P00033) 5 Insulin/IGF pathway-mitogen activated protein kinase kinase/MAP kinase cascade 5 (P00032) Dopamine receptor mediated signaling pathway (P05912) 5 Pyrimidine Metabolism (P02771) 5 Endothelin signaling

pathway (P00019) 5 Blood coagulation (P00011) 5 Adrenaline and noradrenaline biosynthesis (P00001) 4 De novo purine biosynthesis (P02738) 4 PI3 kinase pathway (P00048) 4 Notch signaling pathway (P00045) 4 Muscarinic acetylcholine receptor 2 and 4 signaling pathway (P00043) 4 Nicotine degradation (P05914) 4 5-Hydroxytryptamine degradation (P04372) 4 Axon guidance mediated by semaphorins (P00007) 3 Alpha adrenergic receptor signaling pathway (P00002) 3 Ubiquitin proteasome pathway (P00060) 3 p53 pathway (P00059) 3 Formyltetrahydroformate biosynthesis (P02743) 3 Plasminogen activating cascade (P00050) 3 Synaptic vesicle trafficking (P05734) 3 Metabotropic glutamate receptor group III pathway (P00039) 3 Ionotropic glutamate receptor pathway (P00037) 3 Thyrotropin-releasing hormone receptor signaling pathway (P04394) 3 2-arachidonoylglycerol biosynthesis (P05726) 3 Serine glycine biosynthesis (P02776) 3 Nicotine pharmacodynamics pathway (P06587) 3 Cytoskeletal regulation by Rho GTPase (P00016) 3 Purine metabolism (P02769) 3 5HT2 type receptor mediated signaling pathway (P04374) 3 Axon guidance mediated by netrin (P00009) 2 Pyridoxal-5-phosphate biosynthesis (P02759) 2 Alzheimer disease-amyloid secretase pathway (P00003) 2 Heme biosynthesis (P02746) 2 VEGF signaling pathway (P00056) 2 Tetrahydrofolate biosynthesis (P02742) 2 De novo pyrimidine ribonucleotides biosynthesis (P02740) 2 Androgen/estrogene/progesterone biosynthesis (P02727) 2 JAK/STAT signaling pathway (P00038) 2 Vitamin B6 metabolism (P02787) 2 p53 pathway by glucose deprivation (P04397) 2 Hypoxia response via HIF activation (P00030) 2 Vitamin D metabolism and pathway (P04396) 2 Ras Pathway (P04393) 2 Oxytocin receptor mediated signaling pathway (P04391) 2 p38 MAPK pathway (P05918) 2 Hedgehog signaling pathway (P00025) 2 Glycolysis (P00024) 2 General transcription by RNA polymerase I (P00022) 2 Pyruvate metabolism (P02772) 2 Histamine H1 receptor mediated signaling pathway (P04385) 2 Corticotropin releasing factor receptor signaling pathway (P04380) 2 Circadian clock system (P00015) 2 5HT4 type receptor mediated signaling pathway (P04376) 2 5HT3 type receptor mediated signaling pathway (P04375) 2 5HT1 type receptor mediated signaling pathway (P04373) 2 Toll pathway-drosophila (P06217) 1 SCW signaling pathway (P06216) 1 GBB signaling pathway (P06214) 1 DPP signaling pathway (P06213) 1 DPP-SCW signaling pathway (P06212) 1 BMP/activin signaling pathway-drosophila (P06211) 1 N-acetylglucosamine metabolism (P02756) 1 Methylmalonyl pathway (P02755) 1 Methionine biosynthesis (P02753) 1 Leucine biosynthesis (P02749) 1 Isoleucine biosynthesis (P02748) 1 mRNA splicing (P00058) 1 Histidine biosynthesis (P02747) 1 Transcription regulation by bZIP transcription factor (P00055) 1 Fructose galactose metabolism (P02744) 1 De novo pyrimidine deoxyribonucleotide biosynthesis (P02739) 1 Metabotropic glutamate receptor group I pathway (P00041) 1 Asparagine and aspartate biosynthesis (P02730) 1 Metabotropic glutamate receptor group II pathway (P00040) 1 GABA-B receptor II signaling (P05731) 1 Alanine biosynthesis (P02724) 1 Interferon-gamma signaling pathway (P00035) 1 Adenine and hypoxanthine salvage pathway (P02723) 1 p53 pathway feedback loops 2 (P04398) 1 Valine biosynthesis (P02785) 1 Threonine biosynthesis (P02781) 1 P53 pathway feedback loops 1 (P04392) 1 Bupropion degradation (P05729) 1 Heterotrimeric G-protein signaling pathway-rod outer segment phototransduction 1 (P00028) Opioid proopiomelanocortin pathway (P05917) 1 Sulfate assimilation (P02778) 1 Opioid prodynorphin pathway (P05916) 1 Succinate to propionate conversion (P02777) 1 Opioid proenkephalin pathway (P05915) 1 General transcription regulation (P00023) 1 Salvage pyrimidine ribonucleotides (P02775) 1 Salvage pyrimidine deoxyribonucleotides (P02774) 1 Angiotensin II-stimulated signaling through G proteins and beta-arrestin (P05911) 1 Pyridoxal phosphate salvage pathway (P02770) 1 DNA replication (P00017) 1 Cholesterol biosynthesis (P00014) 1 Beta3 adrenergic receptor signaling pathway (P04379) 1 Beta2 adrenergic receptor signaling pathway (P04378) 1 Beta1 adrenergic receptor signaling pathway (P04377) 1

(92) TABLE-US-00002 TABLE 2 List of amyloid pathway genes being differentially expressed
 Gene id log.sub.2 Fold Change A2M -1.13 ADAM9 1.53 APBB3 0.98 APH1B 2.06 APOL1 -1.66

ATP2A3 1.89 CAPN1 2.13 FAS -1.98 FSTL1 0.96 ITPR3 2.22 LPL 2.74 MMP19 3.21 MMP2 2.76 MMP7 5.75 NDUFA6 -1.57 NDUFB1 -0.78 NDUFB3 -0.75 NDUFV3 -0.86 NOTCH3 2.66 TCF3 1.34 TTR -2.11

Genes Involved in Amyloid Precursor Protein (APP) Metabolism were Differentially Expressed in BA Liver

(93) Pathway analysis for differentially expressed genes (1558) were done using PANTHER (website: www.pantherdb.org/), KEGG Mapper (<https://www.genome.jp/kegg/mapper.html>), DAVID (website: david.ncifcrf.gov/) and also from literatures. Among the list of pathways (Table 1), apart from hormonal and metabolic pathways, few disease related pathways were observed. Alzheimer disease presenilin and amyloid secretase pathway genes (10 nos.) were observed from panther pathways. KEGG pathway analysis also revealed few more genes to be involved in amyloid pathway genes from whole liver RNA-seq analysis. There were a total of 21 genes from total differentially expressed genes involved in amyloid pathway. The list of 21 genes along with log 2FoldChange are displayed in Table 2. Among which, MMP7 (Matrilysin, involved in amyloid-presenilin pathway) being the highest upregulated gene (log 2FoldChange=5.752) among the total (1558) differentially expressed genes.

(94) Expression of APP and its Processed Forms in BA Liver

(95) Immuno-fluorescence analysis for APP on liver sections of BA (n=18) and control (HB, n=5; CC, n=13; CS, n=2) patients revealed elevated expression of APP and/or its processed forms in the bile ducts of all the BA livers but not in all the control livers (FIG. 2).

(96) All patents, patent applications, and other publications, including GenBank Accession Numbers or equivalent sequence identification numbers, cited in this application are incorporated by reference in the entirety of their contents for all purposes.

(97) TABLE-US-00003 TABLE 3 APP_transcripts_final Number of Ensembl_protein_id/
Fasta_Formatted_DNA_sequences_with_cufflinks_variants UniProt_id transcript_ids_and_corres-
ponding_Ensembl_ids 1 ENSP00000284981 >TCONS_00180767|ENST00000346798 (SEQ
ID NO: 1) GGTACCCACTGATGGTAATGCTGGCCTGCTGGCTGAACC
CCAGATTGCCATGTTCTGTGGCAGACTGAACATGCACAT
GAATGTCCAGAATGGGAAGTGGGATTCAGATCCATCAGG
GACCAAAACCTGCATTGATACCAAGGAAGGCATCCTGCA
GTATTGCCAAGAAGTCTACCCTGAACTGCAGATCACCAAT
GTGGTAGAAGCCAACCAACCAGTGACCATCCAGAACTGG
TGCAAGCGGGGCCGCAAGCAGTGCAAGACCCATCCCCA
CTTTGTGATTCCCTACCGCTGCTTAGTTGGTGAGTTTGTA
AGTGATGCCCTTCTCGTTCCTGACAAGTGCAAATTCTTAC
ACCAGGAGAGGATGGATGTTTGCGAACTCATCTTCACT
GGCACACCGTCGCCAAAGAGACATGCAGTGAGAAGAGTA
CCAACCTTGCATGACTACGGCATGTTGCTGCCCTGCGGAA
TTGACAAGTTCCGAGGGGTAGAGTTTGTGTGTTGCCCAC
TGGCTGAAGAAAGTGACAATGTGGATTCTGCTGATGCGG
AGGAGGATGACTCGGATGTCTGGTGGGGCGGAGCAGAC
ACAGACTATGCAGATGGGAGTGAAGACAAAGTAGTAGAA
GTAGCAGAGGAGGAAGAAGTGGCTGAGGTGGAAGAAGA
AGAAGCCGATGATGACGAGGACGATGAGGATGGTGATGA
GGTAGAGGAAGAGGCTGAGGAACCCTACGAAGAAGCCA
CAGAGAGAACCACCAGCATTGCCACCACCACCACCA
CCACAGAGTCTGTGGAAGAGGTGGTTCGAGAGGTGTGCT
CTGAACAAGCCGAGACGGGGCCGTGCCGAGCAATGATC
TCCCGCTGGTACTTTGATGTGACTGAAGGGAAGTGTGCC
CCATTCTTTTACGGCGGATGTGGCGGCAACCGGAACAAC

TTTGACACAGAAAGTTTACTCAAGACTACCCAGGAACCTC
GCCATGTCCCAAAGTTTACTCAAGACTACCCAGGAACCTC
TTGCCCCGAGATCCTGTAAACTTCCTACAACAGCAGCCAG
TACCCCTGATGCCGTTGACAAGTATCTCGAGACACCTGG
GGATGAGAATGAACATGCCCATTTCAGAAAGCCAAAGA
GAGGCTTGAGGCCAAGCACCGAGAGAGAATGTCCCAGG
TCATGAGAGAATGGGAAGAGGCAGAACGTCAAGCAAAGA
ACTTGCCTAAAGCTGATAAGAAGGCAGTTATCCAGCATT
CCAGGAGAAAGTGGAATCTTTGGAACAGGAAGCAGCCAA
CGAGAGACAGCAGCTGGTGGAGACACACATGGCCAGAG
TGGAAGCCATGCTCAATGACCGCCGCCGCTGGCCCTG
GAGAACTACATCACCGCTCTGCAGGCTGTTCTCTCTCGG
CCTCGTCACGTGTTCAATATGCTAAAGAAGTATGTCCGCG
CAGAACAGAAGGACAGACAGCACACCCTAAAGCATTTCG
AGCATGTGCGCATGGTGGATCCCAAGAAAGCCGCTCAGA
TCCGGTCCCAGGTTATGACACACCTCCGTGTGATTTATGA
GCGCATGAATCAGTCTCTCTCCCTGCTCTACAACGTGCCT
GCAGTGGCCGAGGAGATTCAGGATGAAGTTGATGAGCTG
CTTCAGAAAGAGCAAAACTATTCAGATGACGTCTTGGCCA
ACATGATTAGTGAACCAAGGATCAGTTACGGAAACGATG
CTCTCATGCCATCTTTGACCGAAACGAAAACCAACCGTGG
AGCTCCTTCCCGTGAATGGAGAGTTCAGCCTGGACGATC
TCCAGCCGTGGCATTCTTTTGGGGCTGACTCTGTGCCAG
CCAACACAGAAAACGAAGTTGAGCCTGTTGATGCCCCGCC
CTGCTGCCGACCGAGGACTGACCACTCGACCAGGTTCTG
GGTTGACAAATATCAAGACGGAGGAGATCTCTGAAGTGA
AGATGGATGCAGAATTCCGACATGACTCAGGATATGAAG
TTCATCATCAAAAATTGGTGTTCTTTGCAGAAGATGTGGG
TTCAAACAAAGGTGCAATCATTGGACTCATGGTGGGCGG
TGTTGTCATAGCGACAGTGATCGTCATCACCTTGGTGATG
CTGAAGAAGAAACAGTACACATCCATTCATCATGGTGTGG
TGGAGGTTGACGCCGCTGTCACCCCAGAGGAGCGCCAC
CTGTCCAAGATGCAGCAGAACGGCTACGAAAATCCAACC
TACAAGTTCTTTGAGCAGATGCAGAACTAGACCCCCGCC
ACAGCAGCCTCTGAAGTTGGACAGCAAAACCATTGCTTC
ACTACCCATCGGTGTCCATTTATAGAATAATGTGGGAAGA
AACAAACCCGTTTTATGATTTACTCATTATCGCCTTTTGAC
AGCTGTGCTGTAACACAAGTAGATGCCTGAACTTGAATTA
ATCCACACATCAGTAATGTATTCTATCTCTCTTTACATTTT
GGTCTCTATACTACATTATTAATGGGTTTTGTGTACTGTAA
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Claims

1. A method for diagnosing biliary atresia (BA) or assessing risk of developing BA in an infant, comprising: (i) determining expression level of amyloid precursor protein (APP) in a liver sample taken from the infant; (ii) detecting an increase in the APP expression level from step (i) when compared with a standard control value; and (iii) determining the infant as having BA or at risk of developing BA.
 2. The method of claim 1, wherein the APP expression level is APP mRNA level.
 3. The method of claim 1, wherein the APP expression level is APP protein level.
 4. The method of claim 2, wherein step (i) comprises a reverse transcription polymerase chain reaction (RT-PCR).
 5. The method of claim 3, wherein step (i) comprises an immunoassay.
 6. The method of claim 1, wherein the liver sample is a liver biopsy.
 7. The method of claim 1, further comprising a step, following step (iii), of performing at least one additional diagnostic test for BA.
 8. The method of claim 7, further comprising a step, following step (iii), of treating the infant with Kasai procedure.
 9. A method for assessing effectiveness of Kasai procedure in an infant who has BA and has undergone Kasai procedure, comprising: (i) determining expression level of amyloid precursor protein (APP) in a liver sample taken from the infant; (ii) detecting an increase in the APP expression level from step (i) when compared with a standard control value; and (iii) determining the Kasai procedure as ineffective.
 10. The method of claim 9, wherein the APP expression level is APP mRNA level.
 11. The method of claim 9, wherein the APP expression level is APP protein level.
 12. The method of claim 10, wherein step (i) comprises an RT-PCR.
 13. The method of claim 11, wherein step (i) comprises an immunoassay.
 14. The method of claim 9, wherein the liver sample is a liver biopsy.
 15. The method of claim 9, further comprising a step, following step (iii), of performing liver transplant.
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